7.4 Improvements for the data producers

Messflix realizes that this ML suddenly takes off, together with its data mesh. The company has a recommendation system running, automatically optimized its ad campaigns, and has a bunch of forecasting systems in place.

The needed data products, however, are pretty intense in the sense that all three teams (Green, Yellow, and Gray) start to complain about the amount of transformation work they have to do. They feel like they don't have the right tools to handle intense and complex data transformations. In particular, they have problems handling a bunch of computations in sequence to compute their datasets.

And they might be right. Data teams deploy data orchestrators, transformers, or parallel computing frameworks to handle these kinds of things. The problem the three teams describe sounds like an orchestration problem. So, our Data team decides to provide a data orchestration tool called Apache Airflow to the data producers.

DEFINITION *Apache Airflow* is an open source platform, a so-called *data orchestrator*, to run, schedule, and monitor computational workloads. It runs on Python and has a UI. Basically, Airflow allows you to schedule a bunch of Python scripts and run them in useful sequences.

The Data team members choose to provide Airflow as a self-service, meaning they would again provide a Terraform template together with an onboarding session. The beauty of the Terraform template is that they are able to include configurations that allow them to directly hook up Airflow to the rest of the data platform.

Providing hosted Airflow vs. Airflow infrastructure template

Our data platform team members choose to provide an Airflow infrastructure template. They do so because all the currently relevant Messflix teams are well educated in DevOps.